



SEE THE FUTURE STARTUP COMPETITION 2026 · COLUMBIA CLIMATE SCHOOL

BetterBarter

A campus marketplace for buying, selling, giving and receiving used things. It is built on four ideas: **convenience, circularity, self-sustenance and safety.**

THE PROBLEM

Every year, predictably, the same thing happens to the same people.

Campus residents in New York City want to buy, sell, give and receive used things. They do it across WhatsApp group chats, Facebook Marketplace, eBay, thrift stores and Amazon Prime, with nothing joining any of it up.

Poor Timing & Fit

Buyers find a cheap item online and then discover the pickup is an hour away. Sellers reach a general audience with no way to find the people four blocks from them.

No Trust or Verification

For most online transactions, community members agree to buy or sell to an unverified stranger, sometimes having to reveal their own personal details.

Waste, Waste, Waste

Buyers may end up paying more for what other community members are selling at a cheaper rate. Sellers donate or discard what they cannot sell on time.

12.1M t

US furniture and furnishings generated in 2018 — 80.1% landfilled (EPA)

17M t

US textiles the same year; 11.3M t straight to landfill (EPA)

1 week

of May carries most of a residential campus's annual discard

Meet the BetterBarter Marketplace.

One place to do all of it, built on convenience, circularity, self-sustenance and safety.

Campus-specific Community

Buy, sell, give, or receive items exclusively from verified campus community members. Filter by distance and choose to transact with people who live near you.

Environmentally Conscious

Two people crossing a campus on foot or by bike. No vehicle and no packaging. Objects get a longer life and stay out of the skip for another year or two.

Built-in Safety

Chat with other members using only your first name. No phone numbers, no emails, no personal information exchanged. Meet up on or near campus, whatever is safe and convenient for you.

Variety & Relevance

Access new or used items from textbooks and furniture to fashion and baby toys, all from people nearby. We are more alike than we think.

Deployed and running today.

List it in two steps

Up to three photos, then a short form: category, condition, size, how you are listing it, and where you want to meet. The meeting places are a fixed list of public spots on campus.

Only your campus sees it

One verified university email per account, enforced in the database itself. Every school gets its own Marketplace, walled off from the others.

Claim, then meet

Claiming opens a thread and holds the item for three hours, so only one person walks to the lobby. Messages containing phone numbers or addresses are refused when you press send.

The community is the logistics

Some things take two people and a trolley. A listing can say so, and another student on the same campus takes the job for a fee they agree between themselves. Still no vehicle.

Both sides confirm, in person

Six digits, three on each phone. Each app only ever sees its own half, so both people have to be standing there for the exchange to count.

It keeps itself tidy

An hourly job asks whether a listing is still there and retires it if nobody answers. That keeps the Marketplace showing things you can actually get.

Realistic and simple: five numbers.

Circularity & self-sustenance

Items diverted from waste — used items given away or sold instead of thrown out or donated externally.

Number of items given for free.

Estimated CO₂e avoided per item, on the published model overleaf.

Safety

Percentage of active users with .edu verification. This sits at 100% by construction, since the database requires one to create an account at all.

Reports per 1,000 transactions, and how many we acted on.

One line on why the first number holds up. An exchange counts when six digits, three from each phone, are put together in person. Every other campus reuse figure we have found is either self-reported or weighed at the back of a truck. Ours requires two people in the same place at the same time.

Three levels, kept apart.

MEASURED

1 object

A confirmed exchange. This is the only number on the slide we observe directly.

ESTIMATED

3.45 kg

Mass kept in use, from a published factor table. The category mix leans towards small things, which keeps the average honest.

ESTIMATED + ADJUSTED

7.2 kg

CO₂e avoided, at a **displacement rate of 0.5**. We assume half of reuse events prevent a purchase and half do not.

The assumption we expect to be challenged, so here it is up front:
a rented object earns no carbon credit in this model at all, since it comes back to its owner.

Sources: EPA *Advancing Sustainable Materials Management* (durable goods, textiles); EPA GHG Emission Factors Hub; EPA Waste Management Hierarchy; NCES IPEDS enrolment; WRAP, *Environmental and Economic Benefits of Re-use*. The factor table, the mix and the arithmetic are one file in an open repository.

The material in play, and the revenue against it.

THE MATERIAL

TAM	≈46M objects	≈333,000 t CO ₂ e
SAM	≈14M objects	≈101,000 t
SOM · yr 3	≈300,000	≈2,160 t

15.4M US undergraduates (NCES) × 3 reusable objects a year, at our published 3.45 kg and 7.2 kg CO₂e per object.

THE REVENUE

TAM	\$58M a year
SAM	\$18M a year
SOM · yr 3	\$375k a year

3,896 US degree-granting institutions (NCES 2022–23) at a \$15k licence; SAM is the ~1,200 residential four-year campuses; SOM is 25 of them. Placements and boosts excluded.

Students pay nothing to move an object.

Transaction Fee

A 1–2% fee on higher-value service categories in later phases: tutoring, babysitting, car washing. **Physical items stay free, whether they are given away or sold.**

Institutional Partnerships & Licensing Fees

The main line. Universities already pay to haul this material away and already have to report on circularity. A licence buys them object-level reuse data they can audit, which no collection programme produces today.

Boosted Listings

A small, affordable flat fee for sellers who wish to boost their listing, especially if it is an urgent sale.

Advertisement Fees

Local businesses reaching a verified campus audience: food and drink, clinical services, and anything else that sits alongside what we do rather than competing with it.

One group, then one school, then one university.

PHASE 0 · NOW

50–100 friends

Columbia SPS, invited by hand. Enough people to find out what breaks while the stakes are still low.

PHASE I · 0–1 MONTH

School of Professional Studies

One school, opened properly, through the channels students already read.

PHASE II · 1–6 MONTHS

Columbia University

University-wide, and through the 2027 move-out. That week is the real test of everything here.

PHASE III · 6 MONTHS ON

Other institutions

A university email domain already maps to a campus in the database, so a new school gets its Marketplace the moment its first student signs in.

Columbia's first-years moved in on 29–30 August and orientation runs to 7 September. Phase 0 is happening the same week this is being submitted.

Where we actually are.

What exists

A live product at betterbarter.vercel.app, open source. Row-level security on every table, a hardening pass after the security advisor flagged gaps, sign-up gated to academic domains, rate limits in the database, a reporting and moderation queue, and a confirmation function that counts each exchange once.

The impact model published in full: factor table, category mix, displacement rate and every source. Two people built and shipped all of it in five weeks.

What does not

No paying customers, no institutional partnership, and no pilot users. Two verified Columbia accounts exist and both of them are ours. We have yet to speak to Columbia Housing, EcoReps or the Office of Sustainability.

We would rather say so plainly. What we are putting forward is a working product and a method that holds up under questioning, and those we do have.

A pilot campus, and people willing to argue with the model.

Agung Nugroho

CO-FOUNDER · TECH AND PRODUCT

Builds and runs the Marketplace, the impact model and the data. Graduate student and Graduate Assistant at Columbia SPS, with a background in customer-lifecycle and CRM marketing at telecommunications scale.

Tessa Wong

CO-FOUNDER · STRATEGY AND COMMUNITY

A sustainable development professional. She began her career as a management consultant at Accenture Singapore, then moved into sustainable development research and consulting, first for the development sector in Cambodia and later for the private sector in Singapore.

The Ask

An introduction to a Columbia housing or sustainability office, a pilot campus for the 2027 move-out, and someone willing to take the displacement assumption apart.